

Week 2 Progress

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Bishai Lab

July 2026

This is a team project

Hogan

- Weather and heat framing
- Goggins baseline challenge (Jan)
- Ages 65–69 and 70–74
- Climate X file and hot-month definitions

Roro

- Hospital Authority stroke aggregates
- Outcome timing and transfer path

Dr Bishai

- Original multi-method plan
- Teamwork: go far together
- Jasmine signal; extend definitions to 2023

Also in the room this week

- Writing early (lit + methods) matters
- Equity lens (Marmot): who bears risk
- Peer heat work: summit, district voice, LegCo

This update credits the team's contributions. Judgments on weather and outcomes are still shared.

What the July meeting changed

Earlier plan	Meeting	Working now
AMI + stroke principal-dx	No admission reasons in general HA file	Stroke aggregates; AMI out of scope from that file
One primary heat model	Explore many methods	Labelled multi-spec panel
Heat metrics	Align with Ren/Wang-style spells	Spells, 2D3N, hot-month options with Hogan

Roro's outcome path (working understanding): first GOPC mention of stroke is a *marker*. The actual stroke and hospitalisation generally come earlier. We ignore later mentions and work back to the true event month.

The question we are preparing

How do **monthly** thermal exposures in Hong Kong, 2013–2023, associate with **stroke event aggregates**, and how stable is that picture across a shared panel of heat, cold, spell, and sensitivity specifications?

Complementary to daily heatwave–mortality work in the lab — not a duplicate. Not a daily DLNM; not AMI from a file without admission reasons.

From January: Hogan's intellectual lead

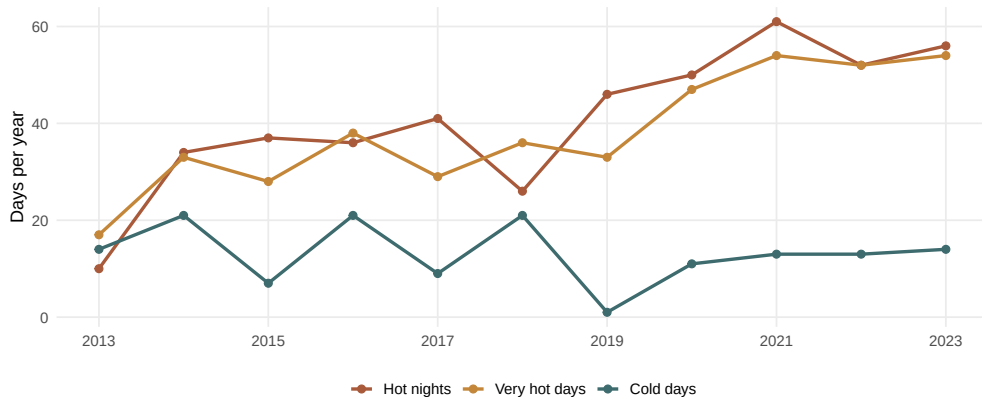
- Pulled Goggins MI (2013) and stroke (2012) as the Hong Kong baseline
- Warned that cold associations are already well documented — novelty must be careful
- Suggested comparing a later period (2013–2023) as an evolutionary question
- Highlighted ages **65–69** and **70–74** as actionable older bands
- Offered to prepare the monthly climate *X* file under Dr Bishai's plan
- Later: define heatwaves, count them by month, flag upper-tail “hot” months (Atmos. Res. heatwave framing; DOI 10.1016/j.atmosres.2024.107845)

Those contributions still structure the weather side. Definitions should be finished **with** Hogan, not around him.

Climate: shared ground truth (descriptive)

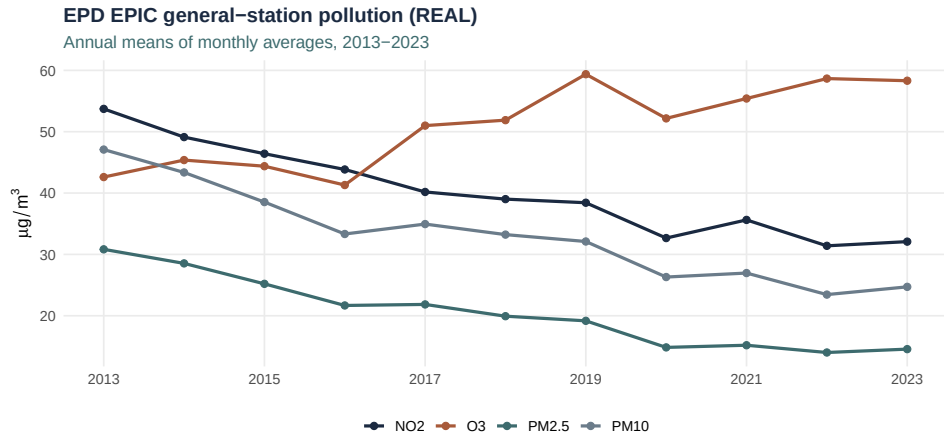
Official thermal extremes at HKO Headquarters

Annual counts, 2013–2023 (environmental data only)



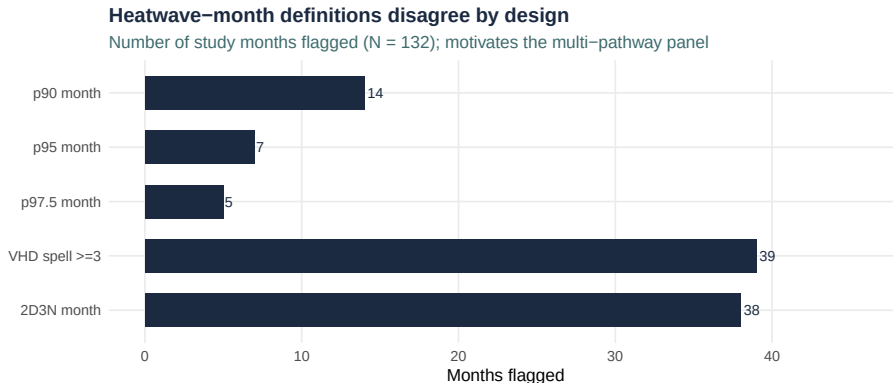
Hot nights and very hot days rose after 2018; cold days persist. Peak hot-night year in-window: **2021 (61)**. Environmental descriptives only — no stroke findings.

Pollution layer (EPD EPIC)



Official monthly general-station means: NO₂, O₃, PM_{2.5}, PM₁₀. NO₂/PM declined; O₃ rose — echoing the January discussion of a changing pollutant mix. Stage pollutants carefully; enter O₃ last.

Why heatwave definitions stay plural



Guo/Gasparrini, Wang/Ren, and the lab's multi-definition mortality work all argue against one universal rule. **Hogan's hot-month idea** (heatwave counts $\rightarrow$ upper-tail months) belongs here as a named option.

A labelled panel — not seventeen discoveries

Family	Examples (lit-anchored)
Continuous T / lag	Tmean; Tmax/Tmin; lag-1
Official extremes	Hot nights, very hot days, cold days (Hogan / HKO)
Spells & 2D3N	Ren/Wang-style burden metrics
Hot-month defs	Percentile months + Hogan count→tail idea
Cold & flu	Cold-primary; influenza co-exposure
Structure	Age bands Hogan flagged; sex if grain allows
Sensitivities	Pollution staging; humidity; COVID window

Headline proposal for Gate 3 *with the team*: continuous Tmax/Tmin + official extreme-day counts.
Full pathway×literature map is in the Laidlaw write-up.

What each person still unlocks

Week 2 readiness: one missing piece

Public layers and analysis plumbing are ready; stroke aggregates unlock inference



Hogan: final weather / heatwave / hot-month / cold-month operationalisation.

Roro: stroke aggregates — first GOPC mention → true event month; dictionary; transfer.

Dr Bishai: confirm which paper is “Jasmine’s” before we call anything a replication.

Public pollution and population layers can support the merge once outcomes arrive.

Honesty line

Ready to discuss

- Environmental descriptives
- Outcome construction logic with Roro
- Weather definitions with Hogan
- Lit + methods writing for Laidlaw

Not ready to claim

- Stroke coefficients
- “Heat replaced cold”
- AMI from this HA file
- Synthetic dry-runs as findings

Literature anchors (selected)

- **Goggins 2012/2013** — cold-dominant HK stroke/AMI baselines (Hogan's challenge)
- **Guo 2024** — hot-night intensity vs official binary flags
- **Wang/Ren 2019** — VHD/HN spells and 2D3N
- **Guo/Gasparrini 2017** — multi-definition heatwaves
- **Yang/Chong 2025** — cold + influenza and stroke
- **Guo 2025** — pollution can amplify temperature–hospitalisation risks

A signal from Dr Bishai — still to verify

Dr Bishai recalls **more non-significant p-values on the hotter side** and **fewer on the colder side** in “Jasmine’s paper.”

- The source paper and exact test universe are not yet confirmed.
- Treat this as a replication prompt, not a result from our stroke data.
- Aim: preserve the earlier definitions and extend the window through 2023.

First: identify “Jasmine’s paper”

Unconfirmed lead	Why it is plausible	Need
Wang / Ren et al. 2019	Hong Kong EHWE mortality, 2006–2015; 5VHD, 5HN, 2D3N	Exact link to “Jasmine”
Liu et al. 2020 meeting-note lead	South China cold/hot attributable burden, 2006–2016	Full citation, DOI, table

A nearby verified paper is Sida Liu et al. (2020), Hong Kong high/low-temperature mortality, 2007–2015. It is an adjacent lead — not assumed to be Jasmine’s.

Definition-harmonised extension to 2023

1. Confirm the paper, outcome, time unit, thresholds, lag and model.
2. Transcribe the original hot/cold rules before opening stroke associations.
3. Check exposure coding in the overlapping years.
4. Apply the unchanged definition through December 2023.
5. Join to Roro's **true stroke-event month**.
6. Report estimates and intervals; treat null-count patterns as secondary.

If the source is daily mortality and ours is monthly stroke, this is definition transport — not an exact replication of its estimand.

Next week: spin out hot and cold months

Hot families

- HKO hot nights / very hot days
- Monthly percentile tails
- VHD, HN and 2D3N spells
- Heatwave counts and event days
- Degree-day intensity
- Anomaly, compound, onset, warnings

Cold families

- HKO cold-day counts
- 12 °C and 10 °C rules
- Monthly percentile tails
- Cold spells and event counts
- Cold degree-days
- Flu, anomaly, warning, onset

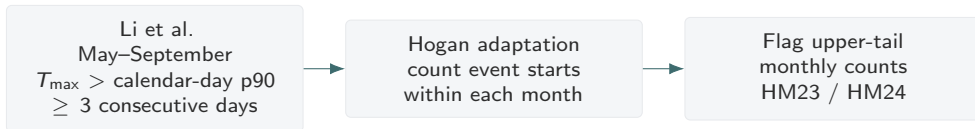
Catalogue: 50 hot-month definitions, 48 cold-month definitions, and paired pathways H01–H12 / C01–C12.

Starter set — small enough to run first

Hot	Rule	Cold	Rule
HM23	Hogan p90 heatwave-count month	CM03	≥ 5 days with $T_{\min} \leq 12^{\circ}\text{C}$
HM08	Monthly mean temperature $\geq \text{p90}$	CM08	Monthly mean temperature $\leq \text{p05}$
HM15	Touches a 5-VHD spell	CM15	Touches a 3-day cold spell
HM17	Touches a 5-HN spell		
HM19	Touches a 2D3N event		

First-wave additions: HM27 night heat-degree burden; HM32 hot + O₃; CM05 any $T_{\min} \leq 10^{\circ}\text{C}$ day; CM30 cold + influenza.

Hogan's atmospheric recipe is one family



Li C et al., *Atmospheric Research* 315 (2025) 107845.

DOI: 10.1016/j.atmosres.2024.107845.

Named and credited — but kept beside HKO counts, Wang/Ren spells, percentiles and cold definitions.

Hypothesis, not mechanism

- Discussion idea: South China populations may appear relatively heat-adapted, while around $\sim 10^{\circ}\text{C}$ may still be consequential.
- This study cannot identify genes, ancestry effects or causal adaptation.
- Housing, air conditioning, behaviour, occupation, age, health care and acclimatisation remain alternative explanations.
- The 10°C rule is a candidate exposure definition, not a biological cutoff.
- No stroke findings yet; synthetic outputs are not findings.

Small ask for next week: confirm Jasmine's citation with Dr Bishai, and lock the weather definitions with Hogan :)

Laidlaw Stage 3 is on the horizon

- Research essay (2000–3000 words) — lit + methods drafts are started
- A0 **portrait** poster (841×1189 mm) — research focus; results pending outcomes
- HKU report form + supervisor endorsement
- Showcase: HKU Laidlaw Society (early Nov 2026) and Annual Conference

Writing early — as other presenters modelled — is how we stay honest about what is ready and what still needs the team.

Thank you

Next small asks — whenever you have a moment :)

- **Dr Bishai:** which paper is “Jasmine’s,” and which table holds the hotter-null / colder-signal pattern?
- **Hogan:** shall we lock the heatwave / hot-month definitions together (including your count→upper-tail idea, and the cold-month starters)?
- **Roro:** when you can, a short note on the first-GOPC→true-stroke timing and the aggregate file path would unlock the merge.

If you want to go fast, go alone; if you want to go far, go together.

Grateful for both of you — and for Dr Bishai’s nudge to listen better.